

Analysis and Experiment of Silicon Photomultiplier

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Abstract

Silicon Photomultiplier tubes are important in multiple areas of medical and particle physics to detect minute amounts of photons. The SiPM converts analog signals of light to digital signals as voltage and produces a gain similar to that of a Photomultiplier Tube (PMT). Multiple characteristics effect the efficiency and accuracy of a SiPM, notably dark count, optical cross talk, and after pulsing. In this experiment, we use the SP5601 LED Driver, SP5600 PSAU, and DT5702 Digitizer to detect possible errors and determine the gain. Significant challenges, such as the accurate measurement of dark counts arising from thermal excitation [1], optical cross-talk where avalanches in one microcell triggered adjacent cells, and photon count to standard deviation curve, were addressed and measured. The results demonstrate that while significant noise must be managed, SiPMs successfully achieve expected gain values of $2.1 \cdot 10^6$ consistent with theoretical predictions and other studies. We additionally saw a lower optical cross talk (OCT) value of 5.6% compared to the 10-20 and a higher dark count rate (DCR) of $1.08 \frac{MHz}{mm^2}$ compared to $0.5 \frac{MHz}{mm^2}$. This experiment provides valuable insights into noise management techniques, necessary for optimizing SiPMs for applications in dark matter physics, quantum physics, and medical imaging.

1 Introduction

Silicon Photomultipliers (SiPMs) are solid-state photodetectors renowned for their capability to detect single photons with high sensitivity (Fig 1) [2]. They function by utilizing the avalanche process within Avalanche Photodiodes (APDs) operated in the Geiger mode, which allows for the multiplication of charge carriers triggered by photon absorption, resulting in a substantial gain of $10^5 - 10^7$ electrons per absorbed photon [3]. This is done by going past the breakdown voltage and allowing for photon avalanches to occur from electron hole pairs. Unlike conventional PMTs, which rely on vacuum-based dynamics, SiPMs leverage silicon's properties for robust and compact photon detection solutions used in multisystem MRI/PET scans as well as WIMP detection in liquid Xenon and Argon [4, 5].

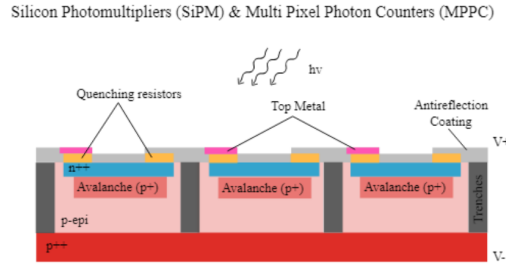


Figure 1: Silicon Photomultiplier Inside

The significance of this experiment lies in its potential to push forward the boundaries of photon detection technology. SiPMs are critically important for applications requiring detailed photon statistics, such as in medical imaging where they can enhance diagnostic clarity, and in particle physics, where they can improve the detection of faint signals in high-energy experiments [4]. Thus, we wish to get an understanding of the precision of the SiPM as well as its time resolution, rise time, and recovery time (Fig 2). By understanding SiPMs' characteristics and limitations, significant improvements can be made to existing systems used in these fields.

The driving motivation for this experiment is the need to thoroughly understand the fundamental limits and noise characteristics of SiPMs to enable their optimal application in various domains. Investigating parameters such as single-photon gain, dark counts, optical cross-talk, and after-pulsing will help refine detector design and usage to enhance performance and reliability.

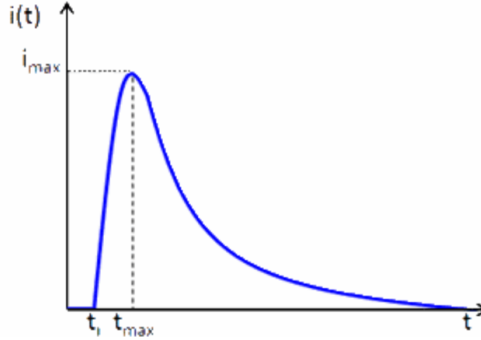


Figure 2: A graph of the current flowing through the terminals of the SiPM. Between t_1 and t_{max} is the rise time, the time resolution is $t_1 - t_f$ and the recovery time is $t_{max} - t_f$

2 Experimental Procedure

2.1 Equipment

The procedure involves connecting the light pulse from the LED driver to the SiPM via optical fibers, with the SiPM output channeled to a digitizer triggered by the LED driver. The setup ensures precise control over voltage settings and signal thresholds, critical for maintaining the integrity of the SiPM's operations, avoiding damage from excessive light or voltage

The following equipment was used for the experiment are as follows and labeled in Fig 3:

- **SP5601 LED Driver** LED Driver that gives the light pulses in different intensities ranging from $10^0 - 10^9$ milliCandelas.
- **SP500 Power Supply and Amplification Unit (PSAU):** Provides the necessary voltage (set to 55 V) and amplifies the signal received from the SiPM.
- **Desktop Digitizer (ADC unit):** Converts the analog signal from the SiPM into digital data for further analysis.
- **Computer with Windows OS:** CAEN HERA is connected to both the PSAU and digitizer, allowing us to control the experiment.

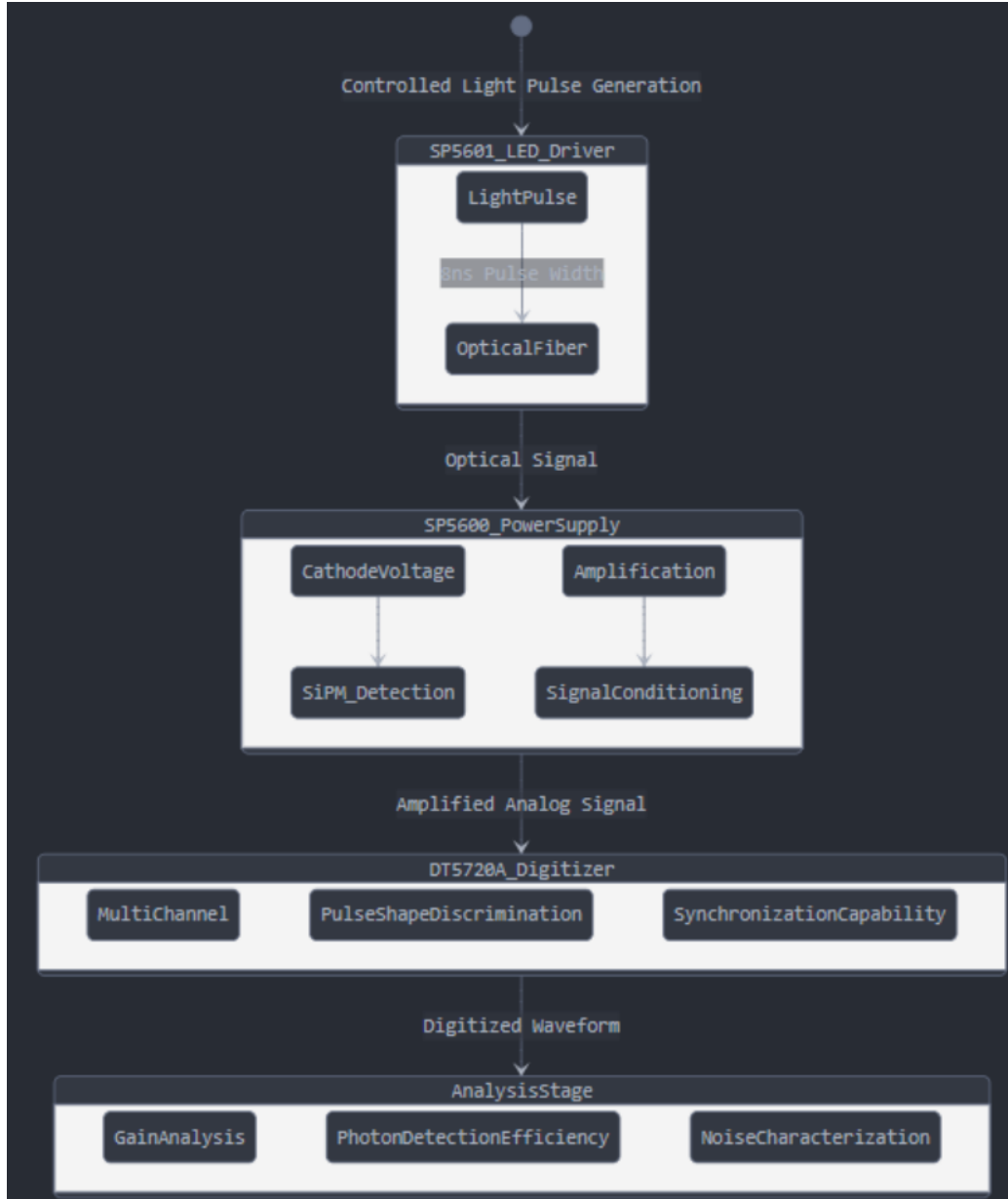


Figure 3: Experimental Setup (Top Down Order): LED Driver, PSAU, SiPM, Desktop Digitizer, Control Software CAEN HERA

2.2 Experimental Setup

The SiPM detects photons, capitalizing on its ability to operate in Geiger mode where a single photon can initiate an avalanche leading to a measurable current pulse. In order to produce accurate results, we proceeded to analyze the light pulse data of three different light intensities: 10, 100, and 1000 milliCandelas. We set the threshold to 70 mV in order to account for the dark and an external voltage of 55 V. Our experiment was ran with 550 thousand counts per light intensity.

Data acquisition involved adjusting the LED driver to emit pulses three different intensities of 10, 100, and 1000, recording the resulting signals, and analyzing spectra through the digitizer.

3 Data and Analysis

3.1 Data Collection

The raw data located in the appendix included the other graphs of data (photon count, Poisson, Sigma squared vs peak, Superposition) under different LED intensities. These measurements were taken with consideration for uncertainty and controlled experimental parameters. In the appendix is a sample of the data for the ADC channels vs photon count graph.

3.2 Gaussian Fit

The histogram data of count vs ADC Channels is then fitted with individual Gaussians. This data will tell us the standard deviation, peak to peak distance, as well as ADC Chanenl values corresponding to photons which is crucial to understand the gain. The Gaussian is modeled as:

$$G(x) = a \cdot \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

For the error propagation, we are using a covariance matrix to calculate the uncertainties of values obtained from fitting data. This uses the

`scipy.optimize.curve_fit` in Python. This contains the following covariance matrix to represent the uncertainties of the amplitude a , mean μ ,

and standard deviation σ .

$$\text{pcov} = \begin{bmatrix} \text{Var}(a) & \text{Cov}(\mu, a) & \text{Cov}(a, \sigma) \\ \text{Cov}(\mu, a) & \text{Var}(\mu) & \text{Cov}(\mu, \sigma) \\ \text{Cov}(\sigma, a) & \text{Cov}(\sigma, \mu) & \text{Var}(\sigma) \end{bmatrix}$$

$$\Delta a = \sqrt{\text{pcov}[0, 0]}, \quad \Delta \mu = \sqrt{\text{pcov}[1, 1]}, \quad \Delta \sigma = \sqrt{\text{pcov}[2, 2]}$$

In Figure 4, the light intensity Gaussian fits are shown below with a fit of each peak:

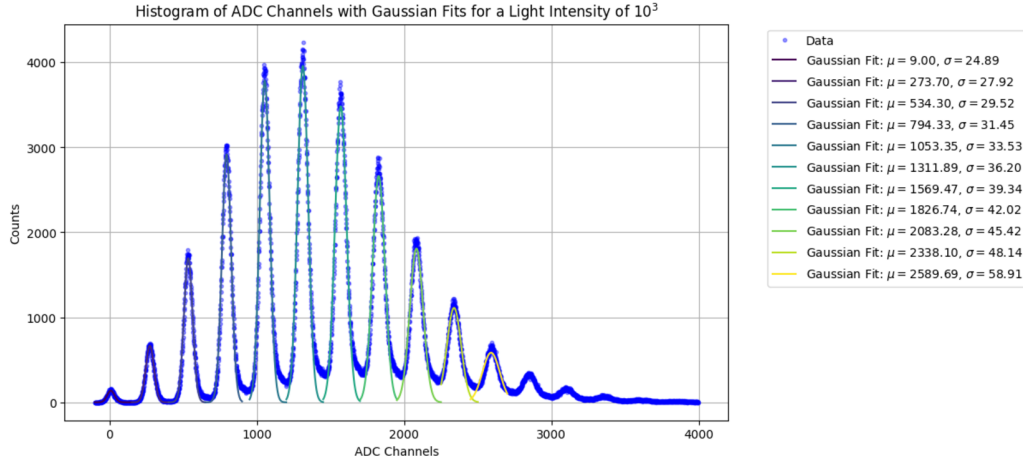


Figure 4: Light Intensity of 10^3 .

3.3 Standard Deviation and Peak Number Correlation

To analyze the relationship between peak number (N) and σ^2 for the 10^3 light intensity data, we'll focus on their linear relationship and apply the Thompson Tau method to improve our analysis.

The data shows a linear relationship between peak number (N) and σ^2 , which can be modeled as:

$$\sigma^2 = \sigma_1^2 N_i + \sigma_0^2$$

where σ_0 is the standard deviation of the zero photon count σ_1 is the first one photon standard deviation.

To refine our analysis, we'll apply the Thompson Tau method to identify and remove outliers which happen to be the 11th and 12th element of our data for a light intensity of 10^3 mCd . This gives us a linear relationship between the two values.

This linear relationship between peak number and σ^2 indicates that the SiPM's response is consistent across different light intensities. Additionally, the SiPM is operating in a regime where each micro cell contributes independently to the overall signal with limited errors.

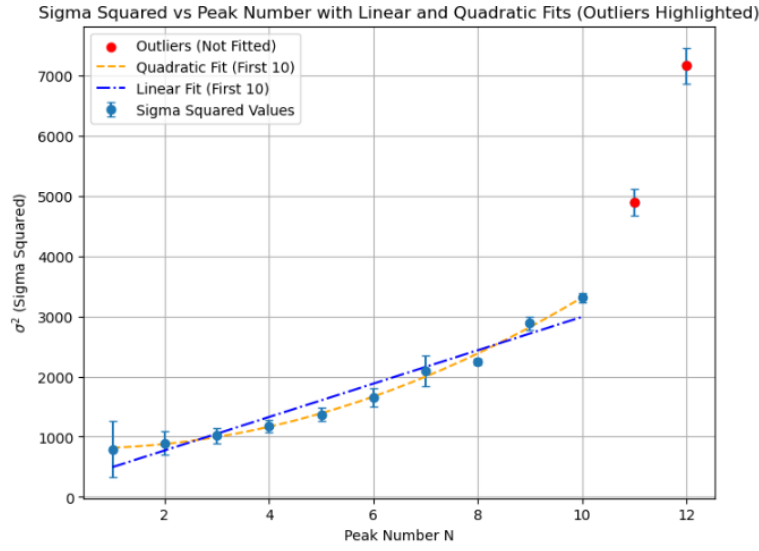


Figure 5: Enter Caption for 10^3 Linear.

3.4 Poisson Distribution

For error propagation in Poisson statistics, the equations can be expressed as:

$$\frac{\partial P_{\text{Poisson},k}}{\partial \lambda} = P_{\text{Poisson},k} \left(\frac{k}{\lambda} - 1 \right)$$

This leads to the uncertainty calculation:

$$\Delta P_{\text{Poisson},k} = P_{\text{Poisson},k} \cdot \left| \frac{k}{\lambda} - 1 \right| \cdot \Delta \lambda$$

The distribution shows two important features:

1. **Relative Similarity:** The expected and observed distributions demonstrate close agreement, except at $k = 0$.
2. **Central Limit Theorem:** The distribution follows the expected behavior according to the Central Limit Theorem.

The zero peak ($k = 0$) in the observed distribution of 10^2 and $10^1 mCd$ shows a slightly higher value than theoretically predicted, which can be attributed to the dark count rate (DCR) of 1.08 Hz/mm^2 measured in the experiment. This elevated zero peak serves as an important indicator of the SiPM's background noise characteristics.

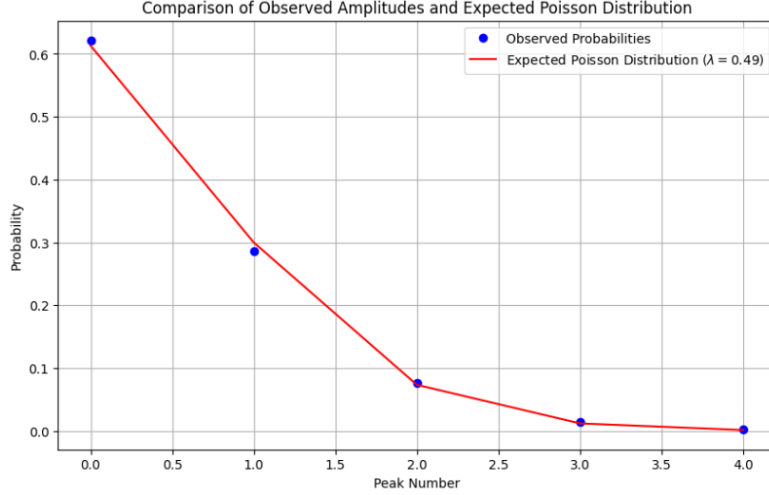


Figure 6: Poisson distribution for 10^1 .

3.5 Gaussian Superposition

The overall spectrum $S(x)$ can be represented as a superposition of Gaussian functions:

$$S(x) = \sum_{N=0}^{N_{\max}} \left[A_N \cdot \exp \left(-\frac{(x - N \cdot g)^2}{2(\sigma_0 \sqrt{N})^2} \right) \right]$$

where:

- A_N represents the amplitude of the N -th peak, proportional to the Poisson probability $P(N, \lambda)$

- g represents the peak-to-peak distance (ΔPP)
- σ_0 is the standard deviation for $N = 1$
- $N_{\max}$ is the maximum number of peaks considered in the analysis

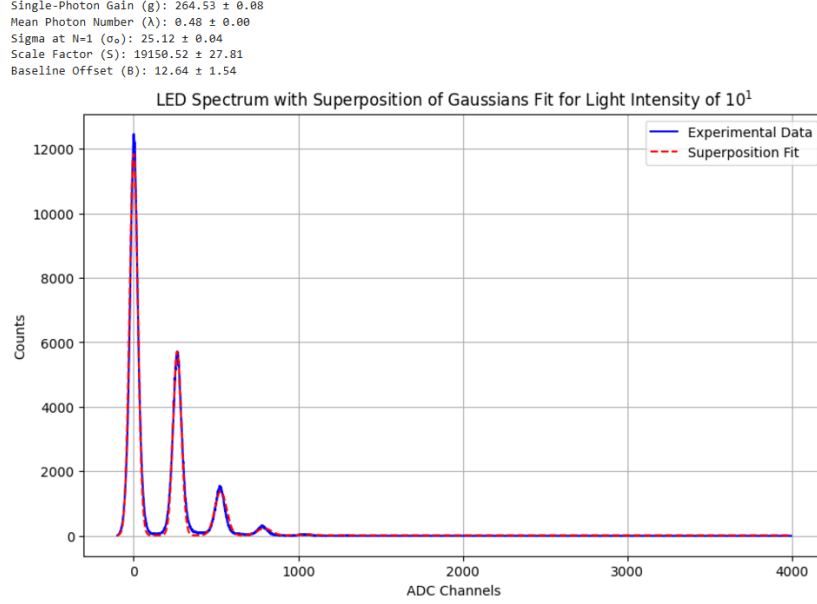


Figure 7: Superposition Gaussian 10^1

As an example, the analysis yielded the following values for a light intensity of $10^1 mCd$. The values are shown in the appendix with the $10^1 mCd$ shown in Fig 7:

- **Peak to peak distance (g):** 264.53 ± 0.08
- **Mean photon number (λ):** 0.48 ± 0.00
- **Standard deviation at $N = 1$ (σ_0):** 25.12 ± 0.04
- **Baseline offset (B):** 19.64 ± 1.54

The spacing between consecutive peaks (g) provides a measure of the single-photon gain by calculating the ADC Channel Conversion Factor $ADC_{c.r.}$.

$$ADC_{c.r.} = \frac{V_{pp} \times \Delta t}{2^{N_{bit}} \times R_{IN} \times G_{PSAU}} \quad (1)$$

Where:

- V_{pp} : Peak-to-peak voltage of the digitizer's dynamic range.
- Δt : Sampling period.
- N_{bit} : Digitizer resolution in bits.
- R_{IN} : Input impedance of the digitizer.
- G_{PSAU} : Gain of the preamplifier.

After we can calculate the gain using the peak to peak separation which is known as g . For convention we change this to ΔPP

$$\text{Gain} = \frac{\Delta PP(\text{ADC}_{ch}) \times \text{ADC}_{c.r.}}{e} \quad (2)$$

Where:

- $\Delta PP(\text{ADC}_{ch})$: Distance between adjacent peaks in ADC channels.
- e : Elementary charge.
- $V_{pp} = 2 \text{ V}$
- $\Delta t = 4 \text{ ns}$
- $N_{bit} = 12$
- $R_{IN} = 50 \Omega$
- $G_{PSAU} = 30$
- $\Delta PP(\text{ADC}_{ch}) = 260 \text{ ADC channels}$

Given the parameters we had, we found the gain to be

$$\text{Gain} = \frac{3.383 \times 10^{-13}}{1.602 \times 10^{-19}} \approx 2.113 \times 10^6 \quad (3)$$

3.6 Dark Count Rate and Optical Cross Talk

The 0.5pe threshold is set to detect signals smaller than one full photoelectron effect and captures the dark count. The DCR measurements show:

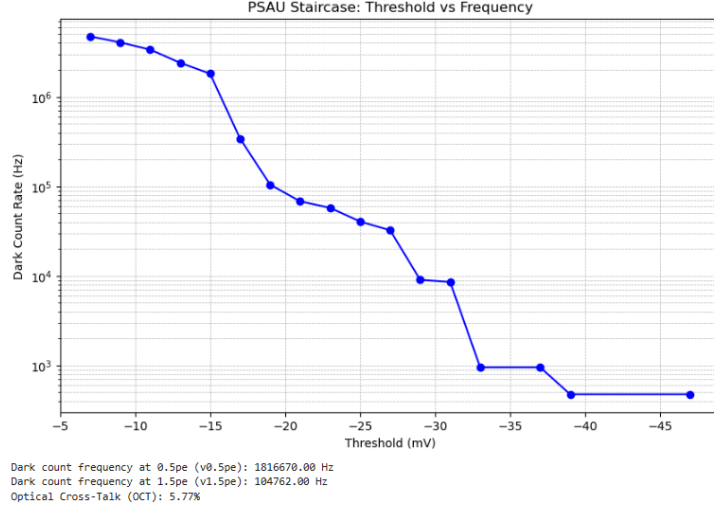


Figure 8: PSAU Staircase

- At 0.5pe threshold: 1.8MHz
- At 1.5pe threshold: 0.10MHz

The normalized dark count rate is calculated by dividing the dark count rate of the entire SiPM by the size of the SiPM which is $1.3 \times 1.3 \text{ mm}^2$. Thus, the

$$\text{DCR} = 1.1 \frac{\text{mHz}}{\text{mm}^2}$$

For the optical cross talk, we are calculating the ratio between the 0.5pe and 1.5pe threshold to determine the number of photons of possible cross-talks. The 1.5 pe threshold is set to detect single photon avalanches.

Using the formula:

$$\text{OCT} = \frac{\text{DCR}_{1.5}}{\text{DCR}_{0.5}} \times 100\%$$

We obtain:

$$\text{OCT} = \frac{0.10}{1.8} \times 100\% = 5.77 \pm 0.02\%$$

4 Discussion

Our measured gain was found to be on the order of 10^6 , which aligns closely with the commonly accepted values for Silicon Photomultipliers (SiPMs) and Photomultiplier Tubes (PMTs) [6]. This consistency suggests that the fundamental avalanche multiplication process within our SiPM closely mirrors that of PMTs, allowing both devices to serve similar roles in photon counting applications. Thus, our measured gain agrees with established literature values, and no significant deviation was observed.

In contrast, the measured Dark Count Rate (DCR) showed a notable discrepancy. While literature values and factory specifications typically report about 0.7 MHz/mm^2 [6], our experiment yielded approximately 1.1 MHz/mm^2 . This difference does not fall within our expected uncertainty range and indicates that additional factors may have influenced our measurement. Potential sources of error include:

- **Thermal Noise:** Although measurements were taken at approximately 25°C , subtle temperature gradients, or insufficient thermal stabilization, could increase the effective DCR.
- **Environmental Factors:** Ambient light leakage, electronic noise coupling, or poor shielding could have artificially elevated the count rate.
- **Device History and Condition:** The SiPM under test may have been aged or stressed by prior usage, affecting its noise characteristics over time. Without detailed knowledge of the device's operational history, we cannot rule out device degradation.

To improve precision and better constrain these uncertainties, a more controlled measurement environment is essential. Strategies for improvement include:

- **Enhanced Temperature Control:** Improving thermal stability through better insulation or a temperature-controlled enclosure to reduce thermally-induced fluctuations in DCR.
- **Refined Bias Voltage Control:** More accurate and stable bias voltage monitoring to ensure the over-voltage setting is well-defined and reproducible.
- **Alternative Optical Cross Talk (OCT) Measurement Meth-**

ods: Employing dual techniques (e.g., time-correlated measurements or alternative threshold methods) to verify OCT results and rule out inconsistencies introduced by the PSAU staircase method.

- **Improved Noise Isolation:** Strengthening electromagnetic shielding and ensuring proper cable management to minimize extraneous noise contributions.

Our OCT measurements, while lower than typical factory specifications, did not severely affect the detector’s operational utility. However, the unexpected lower OCT values might indicate suboptimal measurement conditions or inaccuracies in the PSAU staircase analysis, underscoring the need for cross-checking OCT evaluations with complementary methods.

4.1 What We Learned

Through this experiment, we gained a deeper understanding of photodetector fundamentals. The following insights were particularly valuable:

- **Practical Detector Characterization:** We learned how to measure and interpret key parameters such as gain, DCR, and OCT, and to recognize how environmental and instrumental factors influence these results.
- **Quantum Optics Applications:** The sensitivity of the SiPM to single photons highlights its utility in quantum optics research. Our measurements, albeit imperfect, demonstrate how SiPMs can resolve photon statistics down to the single-photon level.
- **Real-World Challenges:** We discovered the complexity of controlling all experimental variables. Issues like unexpected noise, temperature fluctuations, and aging components reminded us that achieving textbook results often requires iterative refinement of the setup.

4.2 Ease of Use and Recommendations

Once we became familiar with the operation of the SiPM, PSAU, digitizer, and LED driver, the equipment proved relatively user-friendly. However, certain improvements could streamline the process and enhance data quality:

- **Better Calibration Tools:** Integrated calibration references for gain and bias voltage would help ensure consistent and accurate measurements.
- **Stabilized Environmental Conditions:** Enhanced thermal and light shielding could reduce uncertainties, especially for DCR and OCT measurements.
- **Additional Measurement Techniques:** Implementing a secondary method for OCT estimation would provide a valuable cross-check, bolstering confidence in the results.

We would recommend this experiment to other groups interested in learning about solid-state photon detectors and their applications. Despite the deviations from accepted values, the hands-on experience was instructive. It offered a realistic glimpse into detector characterization, quantum-level measurements, and the importance of methodical error analysis.

In conclusion, while our DCR and OCT measurements did not fully align with nominal factory values, these discrepancies led to a better understanding of the conditions that influence SiPM performance. The insights gained here will guide future improvements in both experimental methodology and equipment utilization.

References

- [1] K. Singhrao. Analysis of the afterpulsing and the dark noise spectrum from the deap-3600 detector. Master’s thesis, University of Alberta, 2015.
- [2] S. Gundacker and A. Heering. The silicon photomultiplier: fundamentals and applications of a modern solid-state photon detector. *Physics in Medicine and Biology*, 65(17):17TR01, 2020.
- [3] V. Arosio et al. An educational kit based on a modular silicon photomultiplier system. *Scientific Reports*, 8:5876, 2018.
- [4] C.S. Levin and E.J. Hoffman. Calculation of positron range and its effect on the fundamental limit of positron emission tomography system spatial resolution. *Physics in Medicine and Biology*, 44(3):781–799, 1999.

- [5] P. Lecoq. Development of new scintillators for medical applications. *Nuclear Instruments and Methods in Physics Research Section A*, 809:130–139, 2016.
- [6] AP Technologies. Introduction to sipm technology. <https://www.aptechnologies.co.uk/support/silicon-photomultipliers/introduction-to-sipm-technology>, Accessed 2024.

A Code Availability

All code is available through Github. You can access the repository at the following link:

<https://github.com/MichaelCai2005/SiPM>

B Sample Data and Graphs

This section provides an overview of the raw data files used in this experiment. All data files can be accessed through the following Google Drive link:

<https://drive.google.com/drive/u/0/folders/1s73PiWHvB17XZbHA33qt-jXfAhW9RIU->

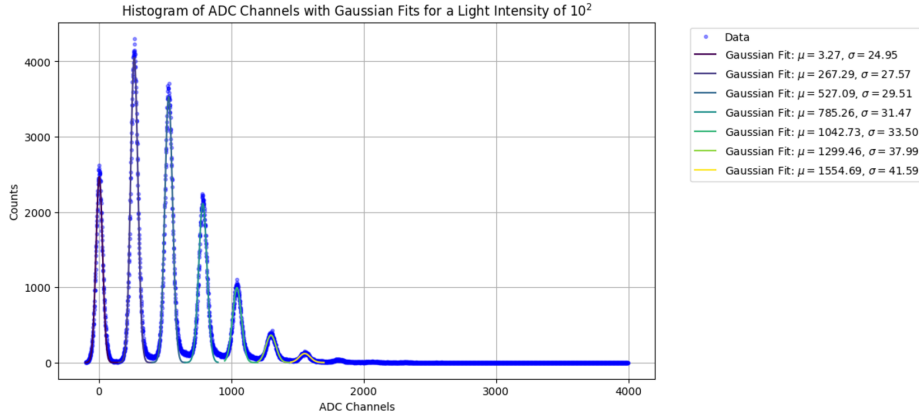


Figure 9: Light Intensity of 10^2 .

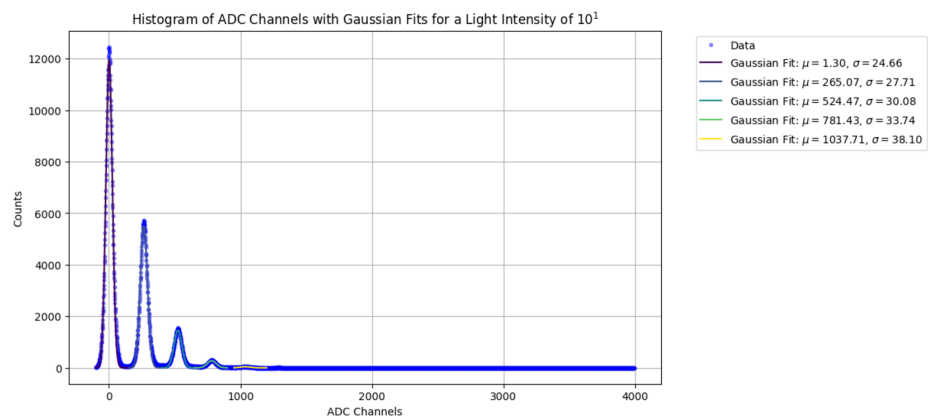


Figure 10: Light Intensity of 10^1 .

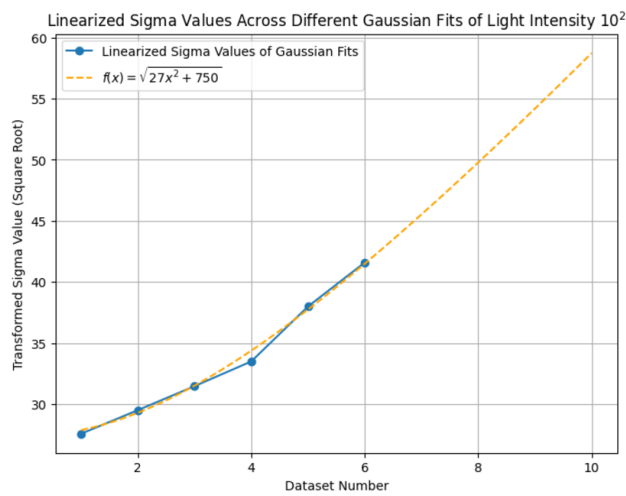


Figure 11: Enter Caption for 10^2 Linear.

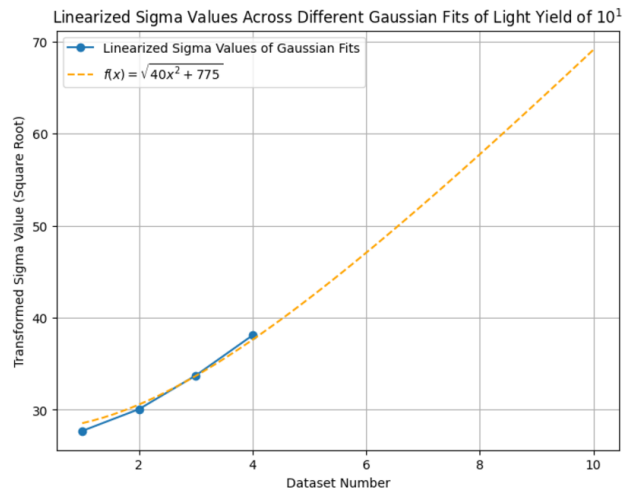


Figure 12: Enter Caption for 10^1 Linear.

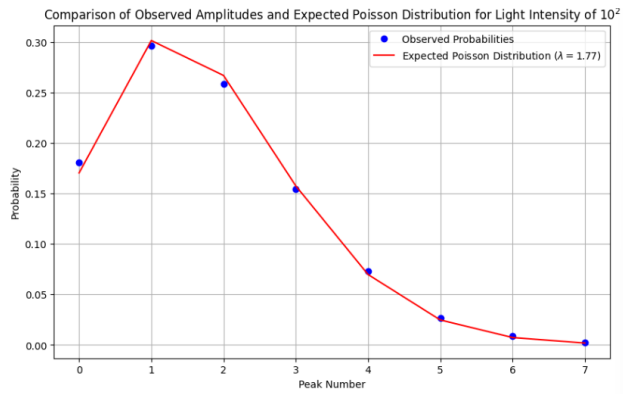


Figure 13: Poisson distribution for 10^2 .

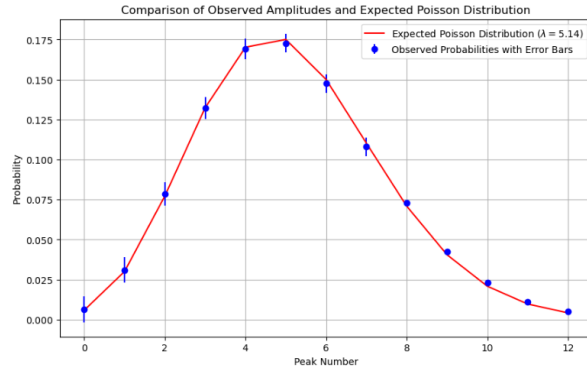


Figure 14: Poisson distribution for 10^3 .

Single-Photon Gain (g): 263.57 ± 0.07
Mean Photon Number (λ): 1.62 ± 0.00
Sigma at N=1 (σ_1): 23.39 ± 0.08
Scale Factor (S): 13067.12 ± 39.60
Baseline Offset (B): 12.83 ± 1.90

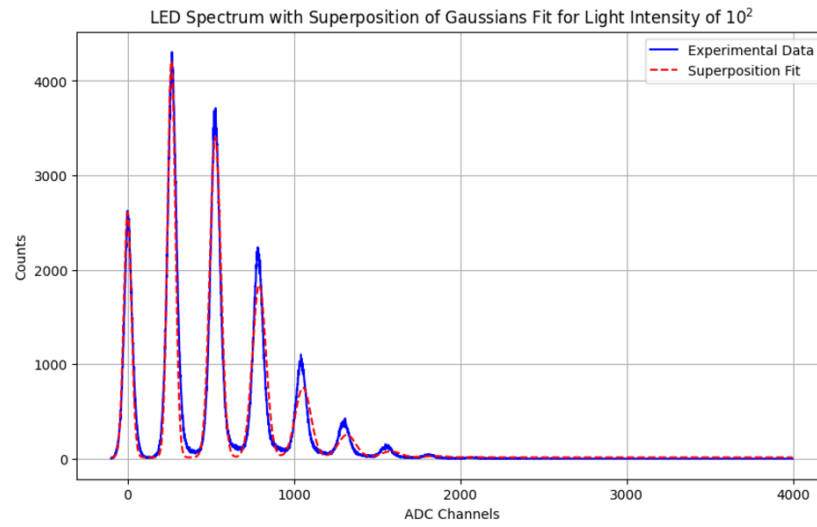


Figure 15: Superposition Gaussian 10^2

Single-Photon Gain (g): 261.96 ± 0.03
Mean Photon Number (λ): 5.19 ± 0.01
Sigma at N=1 (σ_0): 16.09 ± 0.07
Scale Factor (S): 22004.75 ± 75.52
Baseline Offset (B): 104.60 ± 3.28

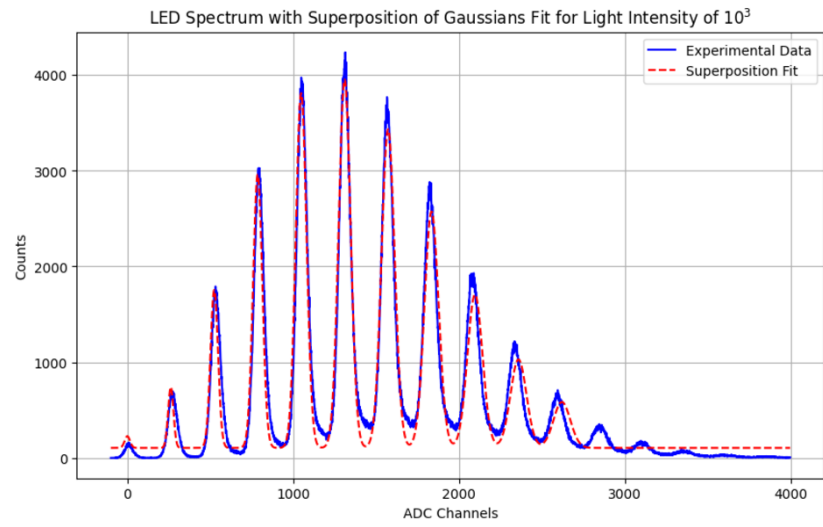


Figure 16: Superposition Gaussian 10^3

ADC Channel	Counts - Charge distribution		
-99.5	0		
-98.5	0		
-97.5	0		
-96.5	0		
-95.5	0		
-94.5	0		
-93.5	0		
-92.5	0		
-91.5	0		
-90.5	0		

(a) The digitizer starts at an ADC count of -100 for the light intensity of 10% milli-Candelas. The data will peak at 0 due to the Gaussian around the 0 photon count.

-8.5	115
-7.5	111
-6.5	126
-5.5	108
-4.5	121
-3.5	146
-2.5	121
-1.5	131
-0.5	144
0.5	114
1.5	122

(b) This showcases the Gaussian centered on the 0 photon data.

Figure 17: Side-by-side graphs of ADC counts and 0-photon Gaussian data.

ADC Channel	Count
127.5	5
128.5	7
129.5	8
130.5	2
131.5	13
132.5	8
133.5	11
134.5	11
135.5	17
136.5	17
137.5	16
138.5	18
139.5	27
140.5	24
141.5	27
142.5	26
143.5	42
144.5	33
145.5	44
146.5	47
147.5	51
148.5	52
149.5	56
150.5	75

Table 1: ADC Channels and Counts Example for Dark Count

ADC Channel	Count
-59.5	136
-58.5	153
-57.5	183
-56.5	158
-55.5	183
-54.5	201
-53.5	221
-52.5	234
-51.5	265
-50.5	288
-49.5	289
-48.5	313
-47.5	357
-46.5	371
-45.5	354
-44.5	395
-43.5	466
-42.5	418
-41.5	519
-40.5	514
-39.5	514
-38.5	568
-37.5	639
-36.5	666
-35.5	709
-34.5	759
-33.5	745
-32.5	817
-31.5	965
-30.5	964

Table 2: ADC Channels and Counts for Light Intensity of 10^2